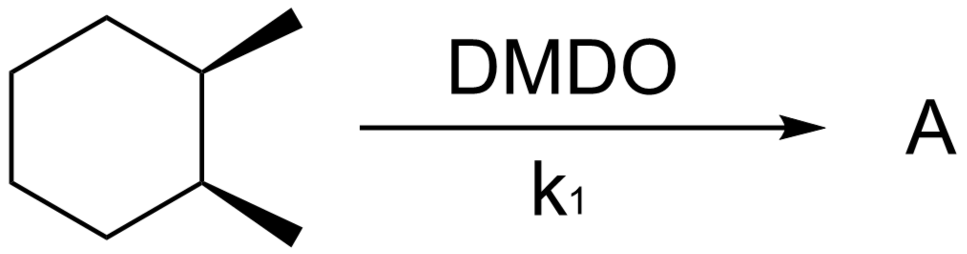
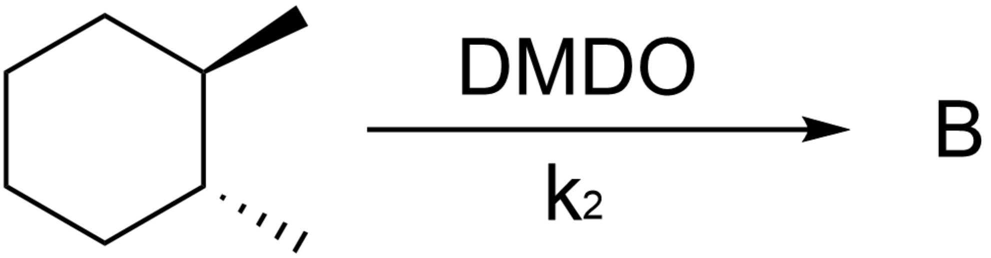
Question



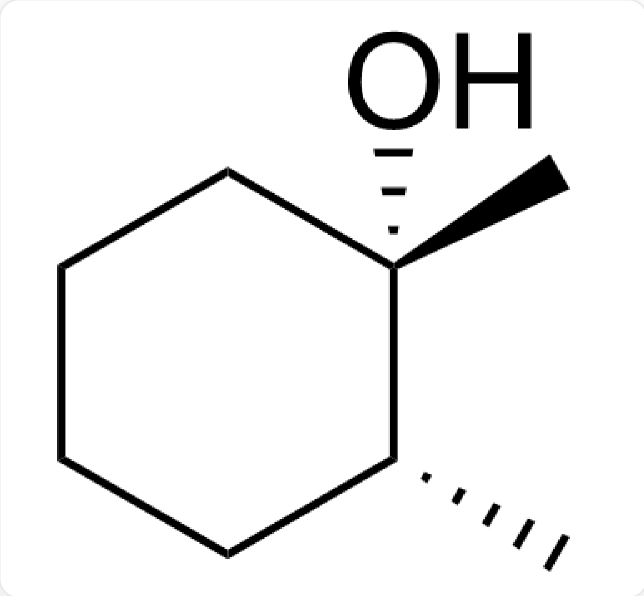
C[C@H]1[C@@H](C)CCCC1>[DMDO]>[A], where A is the reaction product with a rate constant of k_1



C[C@H]1[C@H](C)CCCC1>[DMDO]>[B], where B is the reaction product with a rate constant of k_2

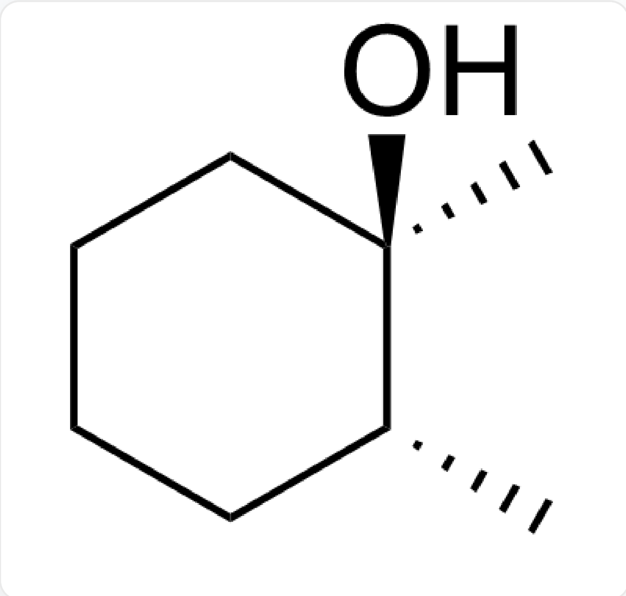
Ignoring enantiomers, provide the structural formulas of products A and B, respectively, and predict the relative magnitudes of k_1 and k_2.

A.



C[C@@]1(O)[C@H](C)CCCC1

Product A

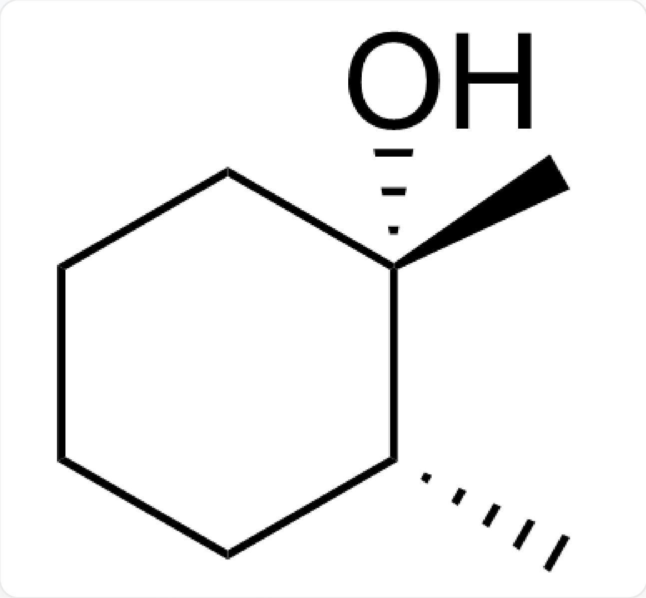


C[C@]1(O)[C@H](C)CCCC1

Product B

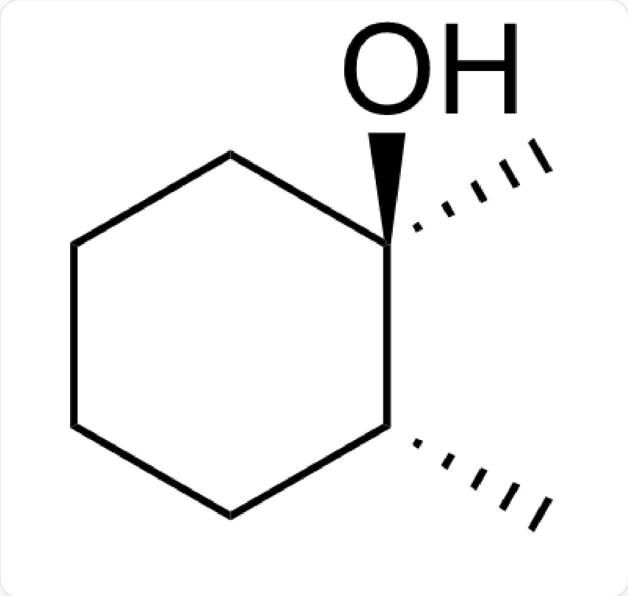
$k_1 > k_2$

B.



C[C@@]1(O)[C@H](C)CCCC1

Product A

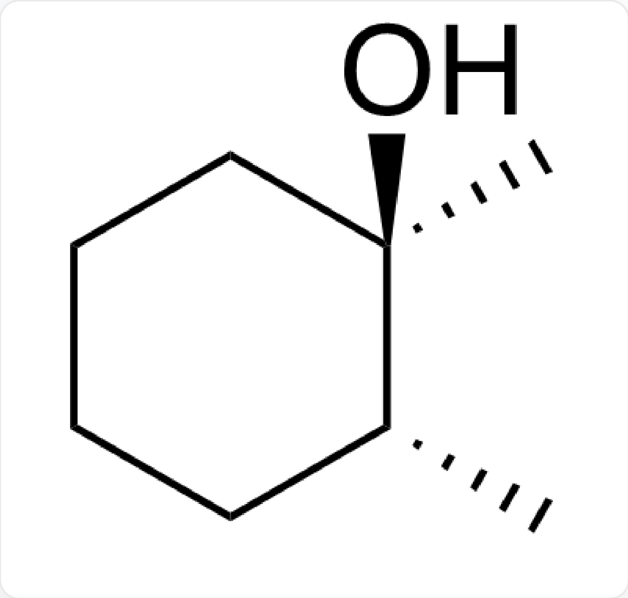


C[C@]1(O)[C@H](C)CCCC1

Product B

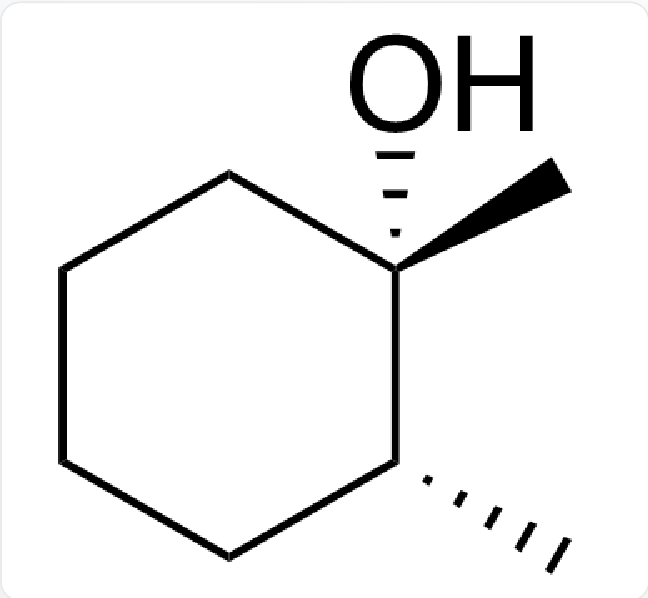
$k_1 < k_2$

C.



C[C@]1(O)[C@H](C)CCCC1

Product A

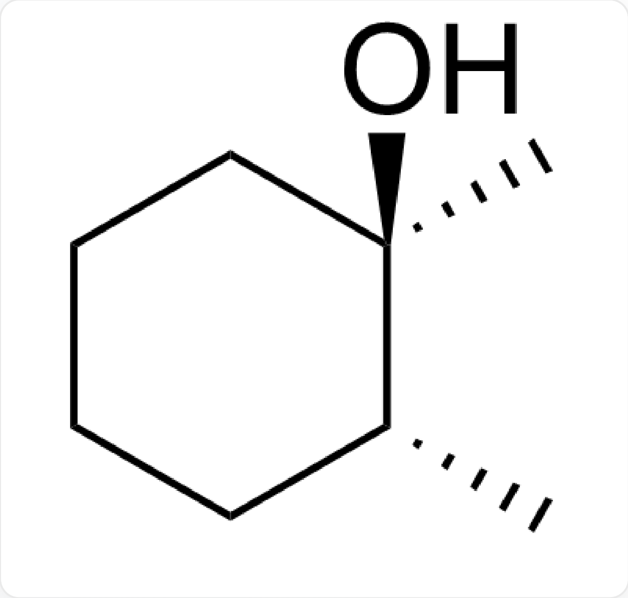


C[C@@]1(O)[C@H](C)CCCC1

Product B

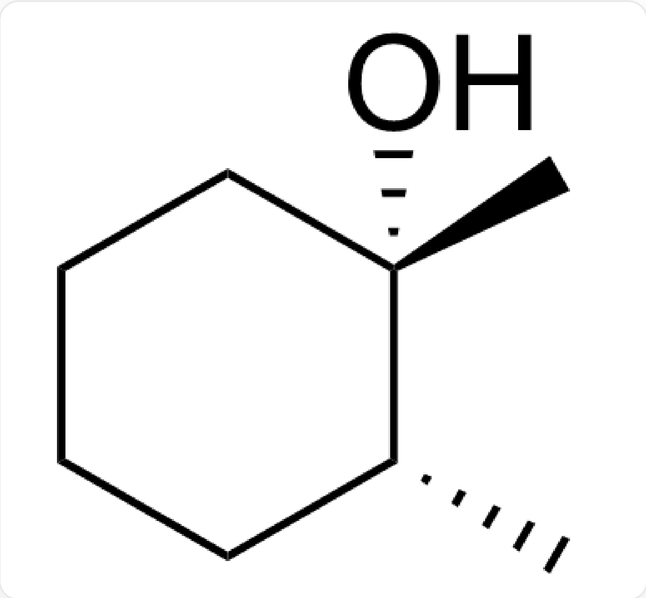
$k_1 > k_2$

D.



C[C@]1(O)[C@H](C)CCCC1

Product A

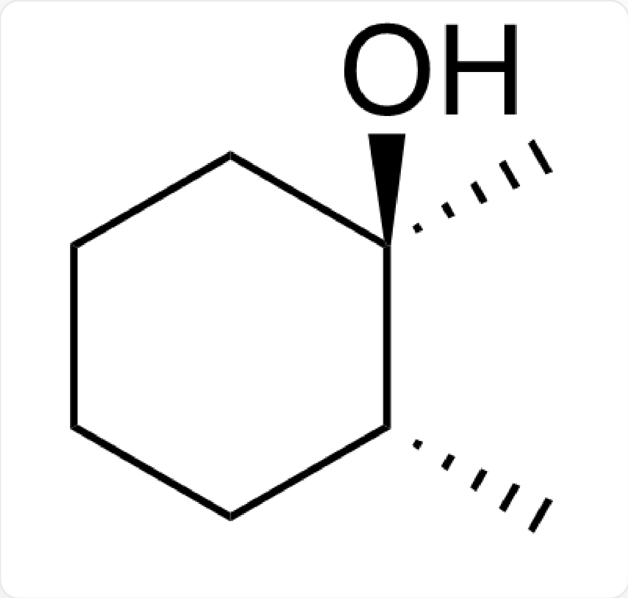


C[C@@]1(O)[C@H](C)CCCC1

Product B

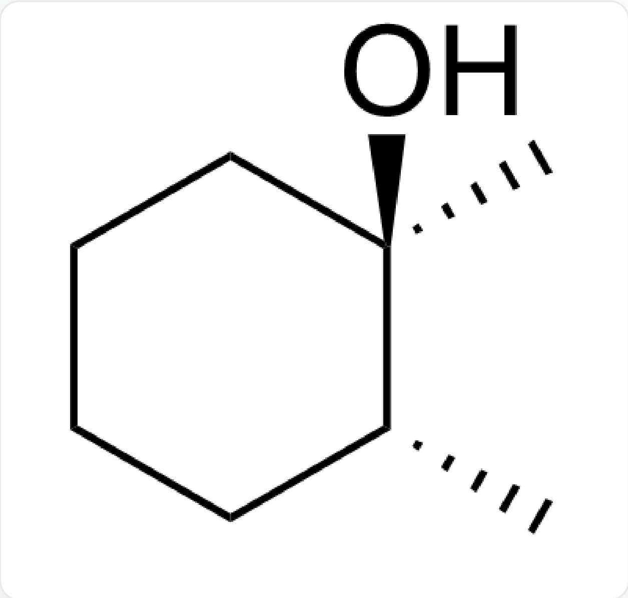
$k_1 < k_2$

E.



C[C@]1(O)[C@H](C)CCCC1

Product A

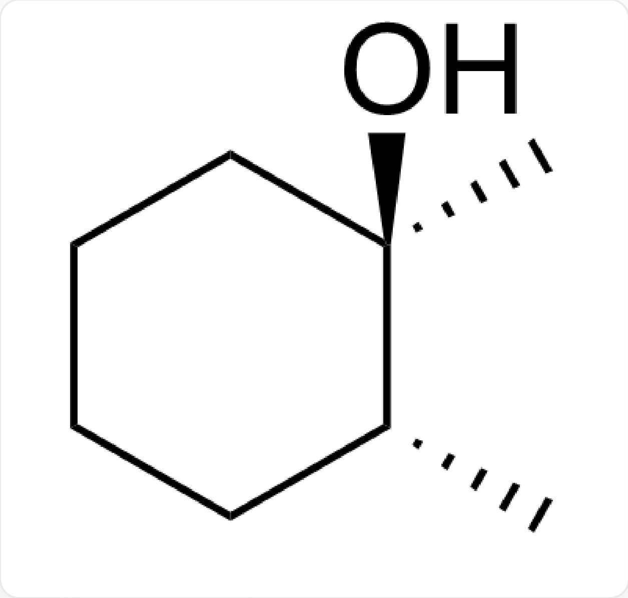


C[C@]1(O)[C@H](C)CCCC1

Product B

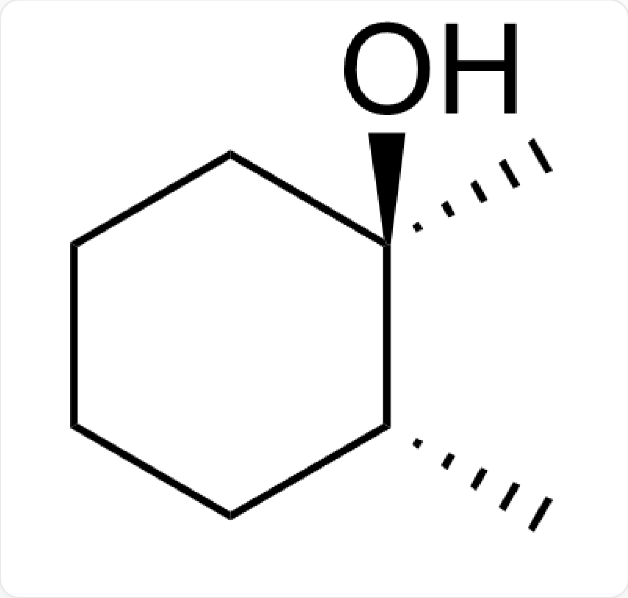
$k_1 > k_2$

F.



C[C@]1(O)[C@H](C)CCCC1

Product A

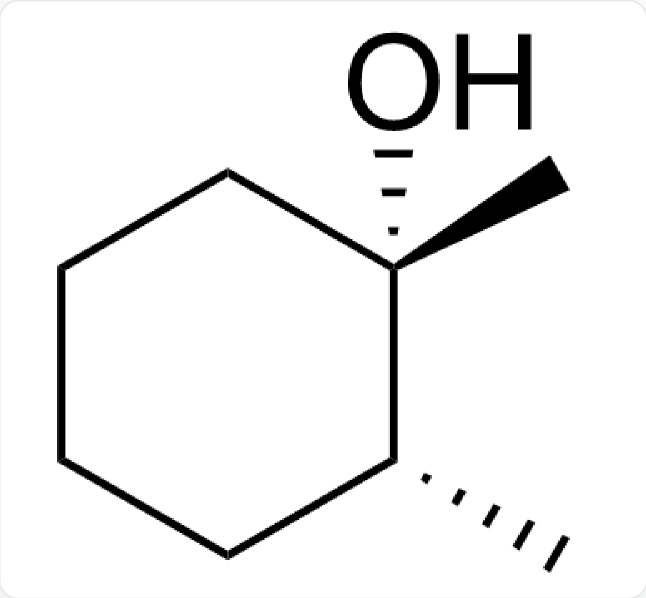


C[C@]1(O)[C@H](C)CCCC1

Product B

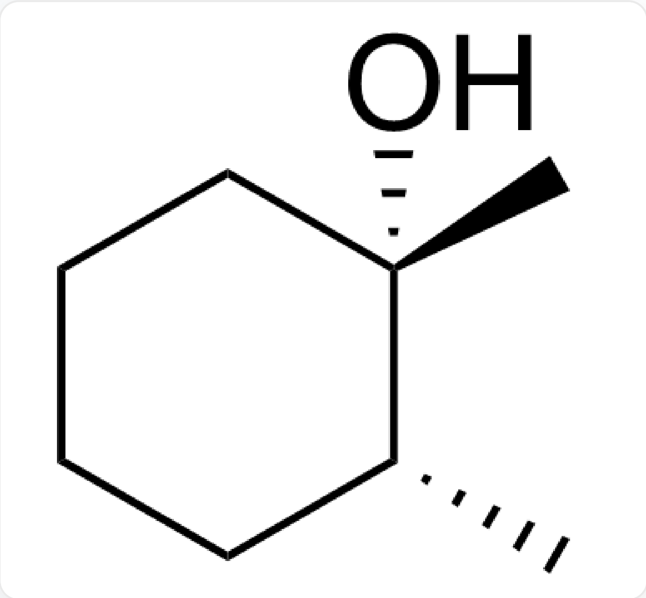
$k_1 < k_2$

G.



C[C@@]1(O)[C@H](C)CCCC1

Product A

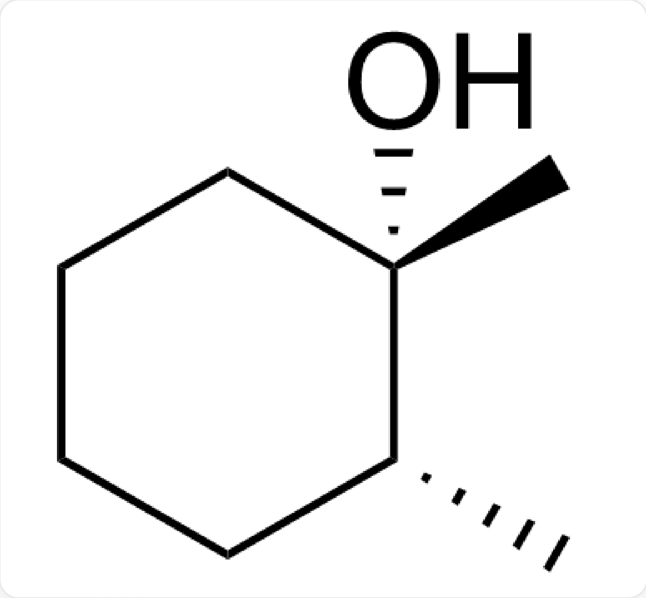


C[C@@]1(O)[C@H](C)CCCC1

Product B

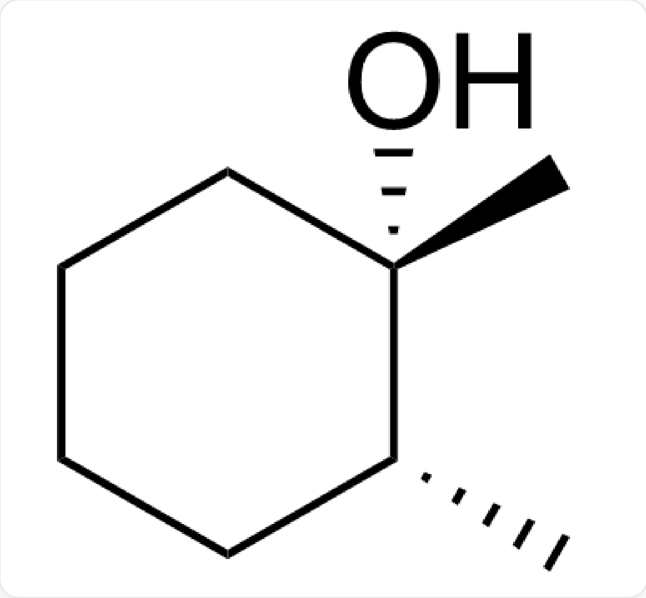
$k_1 > k_2$

H.



C[C@@]1(O)[C@H](C)CCCC1

Product A



C[C@@]1(O)[C@H](C)CCCC1

Product B

$k_1 < k_2$

Answer

Correct Answer: C

Detailed Explanation

The stereoselectivity of this radical reaction is entirely kinetically controlled.

CHECKPOINT 1 PTS

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The reaction first involves hydrogen abstraction by the peroxide, forming a tertiary carbon radical.

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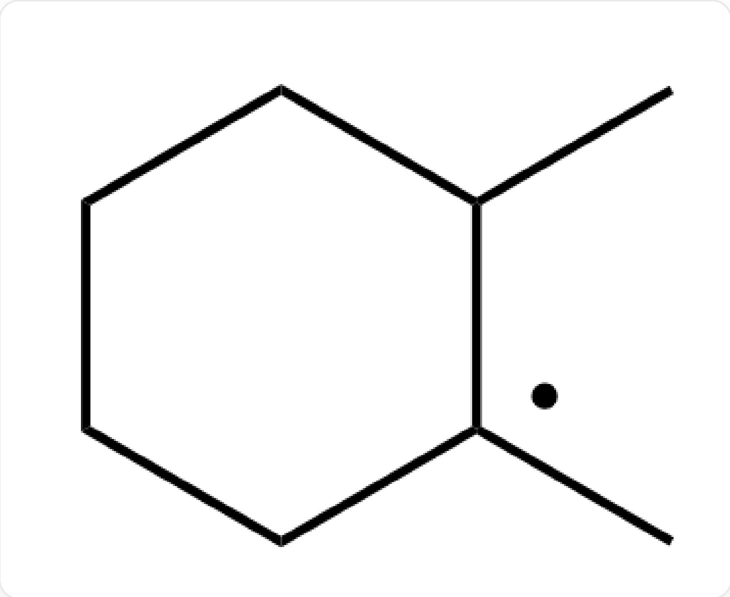

CC1[C](C)CCCC1

Due to the presence of tight radical pairs, the conformation remains unchanged before and after the reaction.

CHECKPOINT 1 PTS

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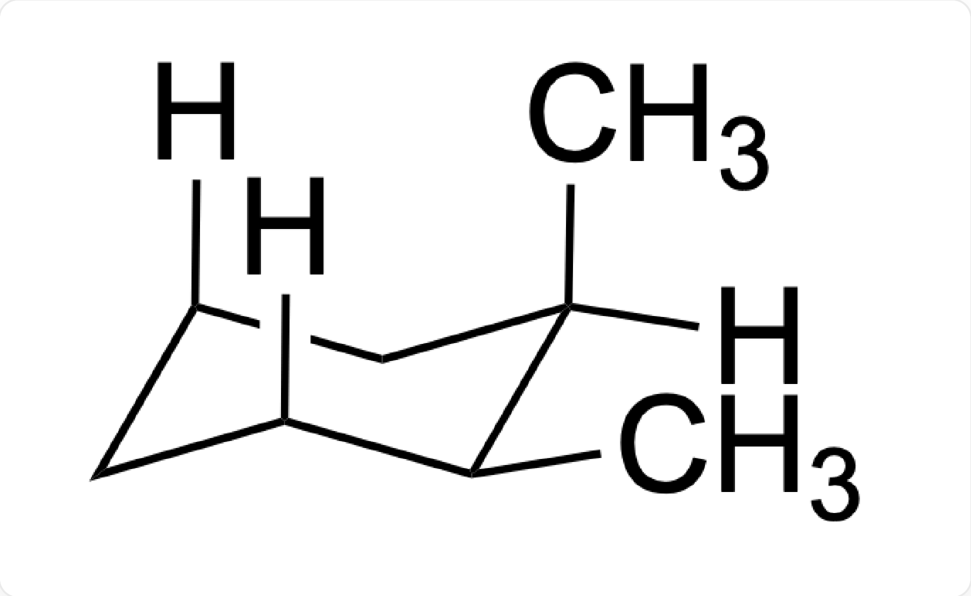
For cis-1,2-dimethylcyclohexane, after hydrogen abstraction, the 1,3-diaxial strain between the axial methyl group and axial hydrogen is relieved, resulting in lower activation energy and faster reaction rate.

CHECKPOINT 1 PTS

For cis-1,2-dimethylcyclohexane, after hydrogen abstraction, the 1,3-diaxial strain between the axial methyl group and axial hydrogen is relieved, resulting in lower activation energy and faster reaction rate



[H]C1CC([H])C[C@](C)([H])[C@H]1C